Models and Techniques Comparison

Rag Techniques used:

Rag technique	What changed	Embedding models used	Top 1	Top 5	MRR	Better or worse than baseline
Baseline dense retrieval	One artifact = one passage, direct FAISS dense retrieval	multilingual-e5-large	0.4889	0.6700	0.5580	default
Structured passages	Rewrote passages in a more explicit structured format before indexing	multilingual-e5-large	0.4844	0.6656	0.5515	worse
Retrieve + rerank(top 10)	Retrieved top-10 candidates, then reranked using lexical overlap	multilingual-e5-large	0.4944	0.6722	0.5621	BEST
Title-boosted retrieval	Repeated the artifact title multiple times inside the indexed passage	multilingual-e5-large	0.4789	0.6700	0.5524	worse
Short-passages retrieval	Used shorter truncated passages instead of full metadata text	multilingual-e5-large	0.4933	0.6522	0.5502	worse
Retrieve + rerank (Top-20)	Retrieved top-20 candidates, then reranked using lexical overlap	multilingual-e5-large	0.4944	0.6722	0.5621	Same as retreive +rerank
Weighted rerank	Retrieved top-10 candidates, then reranked using a weighted combination of dense score and lexical overla	multilingual-e5-large	0.4944	0.6722	0.5621	Same results as retreive + rerank
Top-5 retrieve + rerank	Retrieved top-5 candidates, then reranked using lexical overlap	multilingual-e5-large	0.4933	0.6700	0.5604	better
Title + first sentence retrieval	Used only the title and the first sentence of each artifact text	multilingual-e5-large	0.4844	0.6500	0.5450	worse
Weighted rerank 50/50	Retrieved top-10 candidates, then reranked using a 50/50 combination of	multilingual-e5-large	0.4944	0.6722	0.5621	better

	dense score and lexical overlap					

Models table:

Model	Model Type	Where It Was Used	Technique / Setup	Top-1	Top-5	MRR	Better or Worse
multilingual-e5-large	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.488	0.670	0.558	original
BAAI/bge-m3	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.473	0.654	0.544	worse
multilingual-e5-base	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.474	0.667	0.547	worse

sentence-transformers/paraphrase-multilingual-mpnet-base-v2	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.450	0.654	0.528	worse
intfloat/e5-large-v2	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.474	0.667	0.5468	worse
BAAI/bge-large-en-v1.5	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.481	0.662	0.5490	worse
sentence-transformers/all-mpnet-base-v2	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.466	0.670	0.5436	worse
gte-large	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.473	0.663	0.5461	worse
intfloat/e5-base-v2	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.468	0.660	0.5393	worse
BAAI/bge-base-en-v1.5	Text embedding / retrieval model	Text retrieval	Baseline dense retrieval	0.498	0.671	0.5621	BEST ONE(some probems)